

Static Games of Incomplete Information

Silvana Krasteva

ECON630
Texas A&M University

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Incomplete Information

- Games of Incomplete Information: players do not know the payoffs of other players.
 - Firms may not know the production costs of competitors.
 - Sellers may not know the valuation of potential buyers.
 - Negotiating parties may not know the outside option of the other players.
 - Auctions: bidders do not know the other bidder's valuation for the object.
- We can conveniently transform a game of *incomplete* info to game of *imperfect* info by introducing a new player (nature):
 - Nature moves and selects the type θ_i from cdf $p_i(\cdot)$.
 - Player i observes his type but knows only $p_j(\cdot|\theta_i)$ about player j .

Example

- One incumbent (I) choosing whether to build a new plant;
 - cost of building: c_H with prob. p and c_L with prob. $1 - p$.
- One entrant (E) choosing whether to enter;
- Simultaneous decision with I privately informed about her type.

		E	
		E	NE
I	B	0, -1	2, 0
	NB	2, 1	3, 0

Figure: High Cost

		E	
		E	NE
I	B	3, -1	5, 0
	NB	2, 1	3, 0

Figure: Low Cost

Example: No dominant strategy for low type

		E	
		<i>E</i>	<i>NE</i>
I	<i>B</i>	0, -1	2, 0
	<i>NB</i>	2, 1	3, 0

Figure: High Cost

		E	
		<i>E</i>	<i>NE</i>
I	<i>B</i>	1.5, -1	3.5, 0
	<i>NB</i>	2, 1	3, 0

Figure: Low Cost

Notation

- $\theta_i \in \Theta_i$ where Θ_i is finite for all i ;
- $\Theta \equiv \times_i \Theta_i$ is the set of all possible types, one for each player;
- $p(\theta_1, \dots, \theta_I)$ probability of $(\theta_1, \dots, \theta_I)$;
- Upon observing θ_i , player i updates his belief about the other player's types according to Bayes' rule:

$$p(\theta_{-i}|\theta_i) = \frac{p(\theta_{-i}, \theta_i)}{p(\theta_i)}$$

- If θ_i 's are independent random variables, then $p(\theta_{-i}|\theta_i) = p(\theta_{-i})$;
- S_i - strategy space for player i ;
- $u_i(s_1, \dots, s_I | \theta_1, \dots, \theta_I)$;

Normal Form/Strategic Representation

- A strategic (normal) form of a game $\Gamma = \{I, \Theta, p(\cdot), S, u\}$ specifies:
 - the number of players I ;
 - a type space Θ_i for each player;
 - a probability distribution p over types;
 - a strategy space S_i for each player (may be type dependent);
 - payoff $u_i(s_1, \dots, s_I | \theta_1, \dots, \theta_I)$ for each player;
- The solution concept is *Bayesian Nash equilibrium*.

Bayesian Nash Equilibrium (BNE)

Definition

A strategy profile $(s_1^*(\theta_1), \dots, s_I^*(\theta_I))$ constitutes a Bayesian Nash Equilibrium (BNE) if for all i and all θ_i ,

$$s_i^*(\theta_i) \in \underset{s_i(\theta_i)}{\operatorname{argmax}} \sum_{\theta_{-i}} p(\theta_{-i}|\theta_i) u_i(s_i(\theta_i), s_{-i}^*(\theta_{-i})|\theta_i, \theta_{-i})$$

- The existence of BNE is an immediate consequence of the existence of Nash Equilibrium.

Example: Cournot with Private Information

- 2 firms: 1, 2;
- Inverse demand $p(q) = a - q$ with $q = q_1 + q_2$;
- Firm 1's marginal cost c is common knowledge;
- Firm 2's marginal cost $c = \{c_H, c_L\}$ is private information to firm 2:
 - c_H with probability p ;
 - c_L with probability $1-p$;
- The two firms play Cournot by simultaneously choosing their quantities.
- BNE specifies a strategy for each player and type $(q_1^*, q_2^*(c_H), q_2^*(c_L))$.

Example: Cournot with Private Information

- Given the conjecture q_1^* , the type c_i with $i = \{L, H\}$ maximizes his profit:

$$\begin{aligned} \max_{q_2} (a - q_1^* - q_2 - c_i)q_2 \\ q_2(q_1^*|c_i) = \frac{a - q_1^* - c_i}{2} \end{aligned} \quad (1)$$

- Thus, the equilibrium will be *separating* (i.e. $q_2^*(c_H) \neq q_2^*(c_L)$).

Example: Cournot with Private Information

- Firm 1 maximizes its expected surplus given $(q_2^*(c_H), q_2^*(c_L))$:

$$\max_{q_1} p(a - q_1 - q_2^*(c_H) - c)q_1 + (1 - p)(a - q_1 - q_2^*(c_L) - c)q_1$$

$$q_1^* = p \frac{a - q_2^*(c_H) - c}{2} + (1 - p) \frac{a - q_2^*(c_L) - c}{2} \quad (2)$$

- Equations 1 and 2 should be simultaneously satisfied in equilibrium.

Example: Cournot with Private Information

BNE

$$q_1^* = \frac{a - 2c + pc_H + (1 - p)c_L}{3}$$

$$q_2^*(c_H) = \frac{a - 2c_H + c}{3} + \frac{1 - p}{6}(c_H - c_L)$$

$$q_2^*(c_L) = \frac{a - 2c_L + c}{3} - \frac{p}{6}(c_H - c_L)$$

- Note that the high cost type produces more than the case in which cost c_H is common knowledge.
- Why do you think this is the case?

BNE as a Motivation for Mixed Strategies

- Mixed strategies in games of complete information are interpreted as players randomizing over their pure strategies.
- Mixing can also be motivated by player's uncertainty about the other player's type;

		Wife	
		F	B
Husband	F	$2, 1 + t_w$	$0, 0$
	B	$0, 0$	$1 + t_h, 2$

Figure: Battle of Sexes

- $t_i \sim U[0, x]$;
- player i has private information about i ;

Deriving the BNE in pure strategies

- Higher t_w corresponds to the wife liking football more.
- Suppose that the wife follows a cut-off strategy:
 - go to ballet if $t_w \leq \tilde{t}_w$;
 - go to football if $t_w > \tilde{t}_w$;
- Similarly we can find $\tilde{t}_h$ for the husband;
- Find the cut-offs such that these strategy pair can be supported as a BNE.

Deriving the BNE in pure strategies

Deriving the BNE in pure strategies

References

- Gibbons- Chapter 3
- Fudenberg- Chapter 6